BX-8: A Clean FPGA Implementation of an MSX / R800-Class Machine in Bluespec SystemVerilog

Independent Systems Architecture Study
`btron-bx8` Technical Note

Implementation: `/Users/tonpa/depot/bitedits/btron/src/emulators/bx8/bsv`
Documentation reference: `/Users/tonpa/depot/bitedits/msx.btron.pro/emulators/fpgaMSX`

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Abstract

BX-8 is the entry-level member of the BTRON BX series of clean FPGA RTL machines. It realises an MSX-compatible peripheral set—Video Display Processor (VDP), Programmable Sound Generator (PSG) and related I/O—driven by an R800-class CPU core, all expressed in synthesizable Bluespec SystemVerilog. The design deliberately stays at the 8-bit scale so that the common BSV infrastructure (bus fabrics, clocking, host interface, peripheral-library discipline) can be validated before being reused by the wider 16-, 32- and 64-bit members of the family. This note describes the architectural intent, the organisation of the BSV sources, the relationship to the reference documentation under `fpgaMSX`, and the place of BX-8 inside the overall BX programme. BX-8 itself has no vector unit; the scalable VEX vector architecture begins with BX-16.

1 Motivation and Place in the Series

The BTRON BX programme treats classic media-processor architectures as living subjects of study. BX-8 occupies the foundational position:

- It is small enough that every major block can be held in a single engineer's head.
- It exercises the same Bluespec methodology, peripheral-library style and FPGA flow that later appear in bx-16, bx-32 and bx-64.
- It preserves the characteristic MSX look-and-feel (VDP, PSG/SCC-style sound, cartridge and joystick ports) while remaining fully open and synthesizable.

The authoritative implementation lives at

`/Users/tonpa/depot/bitedits/btron/src/emulators/bx8/bsv`

and the accompanying documentation and reference material for the broader FPGA MSX effort is collected under

`/Users/tonpa/depot/bitedits/msx.btron.pro/emulators/fpgaMSX`.

2 Architectural Outline

2.1 CPU Core

The computational heart is an R800-compatible (or clean re-implementation) 8-bit processor. The R800, originally designed by ASCII Corporation for the MSX turbo R, is a high-performance Z80-family core with additional instructions, a multiplier and improved pipeline behaviour. In BX-8 the core is expressed in BSV so that its interface to the memory and I/O fabric is precise, typed and free of the ambiguities that often accompany hand-written Verilog CPU models.

2.2 Memory and Bus Fabric

A simple but clean multi-master fabric connects:

- the R800 core,
- the Video Display Processor,
- the sound subsystem,
- cartridge / ROM slots,
- system RAM,
- a host or debug interface (UART, JTAG or memory-mapped window).

The fabric is deliberately kept minimal so that latency and arbitration behaviour remain predictable—an essential property when the same infrastructure is later scaled to dual-core and vector designs.

2.3 Video Display Processor (VDP)

The VDP block realises the classic MSX video architecture (TMS9918-compatible modes together with the later V9938/V9958 extensions where appropriate). In BSV the VDP is decomposed into:

- a register file and command decoder,
- a timing generator (horizontal/vertical counters, blanking, sync),
- a sprite engine,
- a background/tile engine,
- a pixel serializer and colour lookup path that feeds a simple VGA or HDMI output stage.

All state transitions are expressed as Bluespec rules, making the interaction between CPU writes, VRAM arbitration and display timing explicit and race-free.

2.4 Sound Subsystem

The sound path supports the characteristic MSX audio devices:

- AY-3-8910-compatible PSG (three tone channels + noise + envelope),
- optional SCC or SCC+ style wavetable channels,
- a simple mixer and sigma-delta or I2S output stage suitable for FPGA boards.

Again the implementation is rule-based; register writes from the CPU and sample generation run concurrently under Bluespec's atomic-rule semantics.

2.5 Peripheral Set

The remaining I/O follows the classic MSX map:

- keyboard matrix and joystick ports,
- memory mapper (where required for larger RAM configurations),
- cartridge slot signalling,
- optional printer or general-purpose parallel port,
- a lightweight debug/host channel for loading ROMs and inspecting state from a workstation.

3 Bluespec SystemVerilog Organisation

The sources under `bx8/bsv` are organised so that each major block is a self-contained package with a clear interface:

```
bx8/
  bsv/
    CPU/ # R800-class core
    VDP/ # Video Display Processor
    Sound/ # PSG / SCC path
    Bus/ # fabric, arbiters, memory map
    Peripherals/ # keyboard, joystick, mapper, ...
    Top/ # system integration, clocking, FPGA pins
```

Key design choices that recur throughout the BX series already appear here:

- Interfaces are explicit BSV interfaces rather than loose collections of wires.
- Timing and concurrency are expressed with rules; the compiler schedules them.
- Parameterisation (RAM size, presence of SCC, video output standard, etc.) is performed at elaboration time.
- A thin Verilog wrapper, generated by the Bluespec compiler, provides the final pin-level interface to the FPGA board.

4 Relationship to the fpgaMSX Documentation

The directory

`/Users/tonpa/depot/bitedits/msx.btron.pro/emulators/fpgaMSX`

holds the broader documentation, test ROMs, timing diagrams and board-support notes that accompany the BX-8 implementation. It serves as the human-readable counterpart to the BSV sources: architectural descriptions, register maps, known deviations from particular MSX models, and synthesis/place-and-route results for the target FPGA families.

Together the two trees form a complete, self-contained package: the BSV sources are the executable specification; the fpgaMSX documentation is the explanatory and validation material.

5 FPGA Implementation Notes

BX-8 is deliberately modest in resource consumption. On a mid-range contemporary FPGA it typically occupies only a fraction of the available LUTs, block RAMs and DSP slices. This economy has two consequences:

1. The design can be synthesised and iterated quickly, making it an ideal test vehicle for the common BX infrastructure.
2. Multiple BX-8 instances, or a BX-8 alongside experimental peripherals, can coexist on a single device—useful for comparative studies and for classroom demonstration.

Clocking is kept simple: a single system clock derived from the board oscillator, with optional secondary clocks for the video pixel domain and the audio sample rate. Reset and start-up sequencing are expressed as ordinary BSV rules, avoiding the ad-hoc initial blocks that often complicate pure Verilog designs.

6 Educational and Methodological Value

Because BX-8 is the smallest member of the series it is the natural starting point for anyone wishing to understand the BX design culture:

- Students can read the entire CPU, VDP and sound packages in a few sittings.
- The same peripheral-library and bus-fabric patterns reappear, only scaled, in bx-16, bx-32 and bx-64.
- Synthesis, simulation and on-hardware debugging can be mastered on a design whose failure modes are still tractable.

In this sense BX-8 is both a useful MSX-class machine in its own right and the pedagogical foundation of the larger programme. The scalable VEX vector architecture is introduced only from BX-16 onward; BX-8 deliberately remains free of vector state so that the common infrastructure can be validated in isolation.

7 Conclusion

BX-8 demonstrates that a classic MSX / R800 peripheral set can be expressed cleanly in Bluespec SystemVerilog and realised on contemporary FPGAs without sacrificing authenticity or readability. By keeping the design compact, well-factored and fully open, it supplies the common infrastructure and the design discipline that the rest of the BTRON BX series builds upon. The sources at `bx8/bsv` together with the documentation under `fpgaMSX` constitute a complete, self-contained realisation of the 8-bit member of the family.

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